CE 334

Assignment 1

The motivation of this study is to highlight the work done to map the figure of earth during last two-three centuries. This mapping will help us to understand the gravitational potential and height differences on terrain points.

→ Key Concept and Methodology:

In 1878, Heinrich Bruns proposed representing Earth's figure as a global polyhedron with gravity potential value at vertices that provides physical height differences. Althought theoretical part works well but practical implementation faced some obstacles that are -

(i) topospheric refraction surely affected the measurement of zenith angle

(ii) Oceans remains inaccessible to geodatic observation. This results in fragmented regional height system with unknown datum offsets.

Methodology -

Brun's polyhedron starts with one triangle of a larger triangular network. Zenith angles at the vertices of the triangles, together with one side length gives the height differences between the vertices. The zenith angls fix the direction of plumb lines relative to the three sides. The shape of triangle is fixed by measured elements, it can still freely shifted and rotated. By this way a whole set of triangles can represent a terrain segment.

→ Importance in Modern geodatic context.
Modern space geodesy revolutionized Bruns vision. The International Terrestial Reference Frame (ITRF), combining VLBI, SLR, GNSS, and DORIS techniques, realizes the geometric components with millimeter precision co-ordinates updated every five years. satellite gravimetry mission provide unprecedented global gravity field knowledge. These combined models integrate satellite gravimetry with terrestrial gravity anomalies, topographic informa Establishing a globally unified height system. addresses critical geodatic challenges. It would reveal systematic distortion in existing hight systems, quantify datum offset between regional networks. Hence a procedure needed to be designed which combines the strength of GPS-leveling over long distances with that of classical leveling over short distances. Finally achieving Bruns 140-year old vision.